

Exercise 1:

DEPARTMENT Scenario

The following requests are made to design a database:

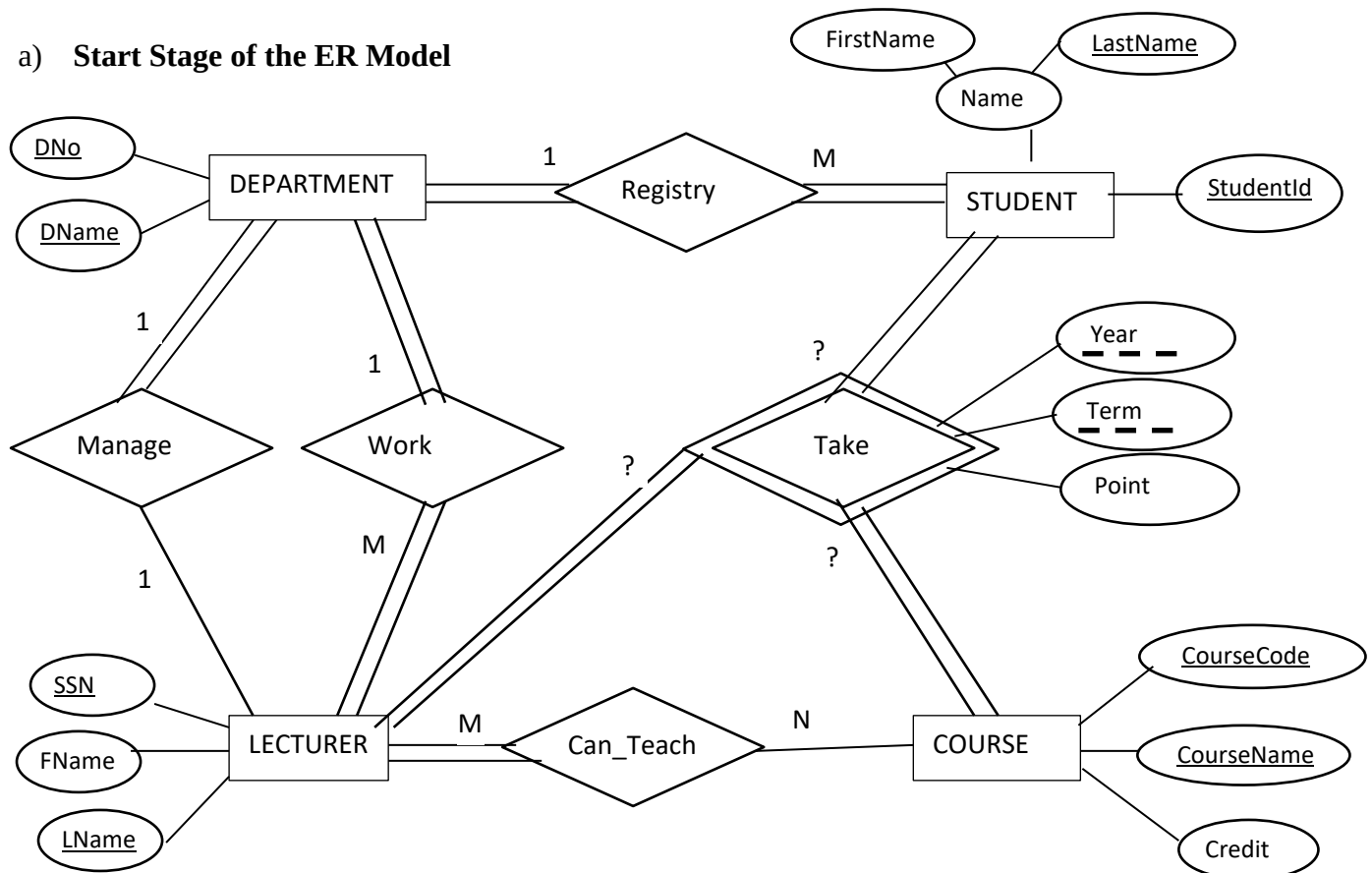
- Department is defined with DNo and DName. Their values can not repeat. It has many students.
- Student is defined with StudentId, Name composed of FirstName and LastName. StudentsId is the identifier. Student is registered in one department only. It has many lecturers.
- Lecturer is defined with SSN(Social Security Number), FName, LName and StartDate. She/He can work in one department only.
- Course is defined with CourseCode, CourseName and Credit. Values of Coursecode and CourseName can not repeat.
- Student takes a course based on Year and Term given by a lecturer and her/his success point is saved with Point at the end of the term.
- Lecturer can give one or more courses and a course can be given by none, one or more lecturers.

According to the requests,

- a) Draw the ER diagram showing attributes, relationships and relationship types and weakness/strogness.
- b) Construct the database schema of Relational Model of ER diagram
- c) Draw the diagram of the schema of Relational Model

SOLUTION:

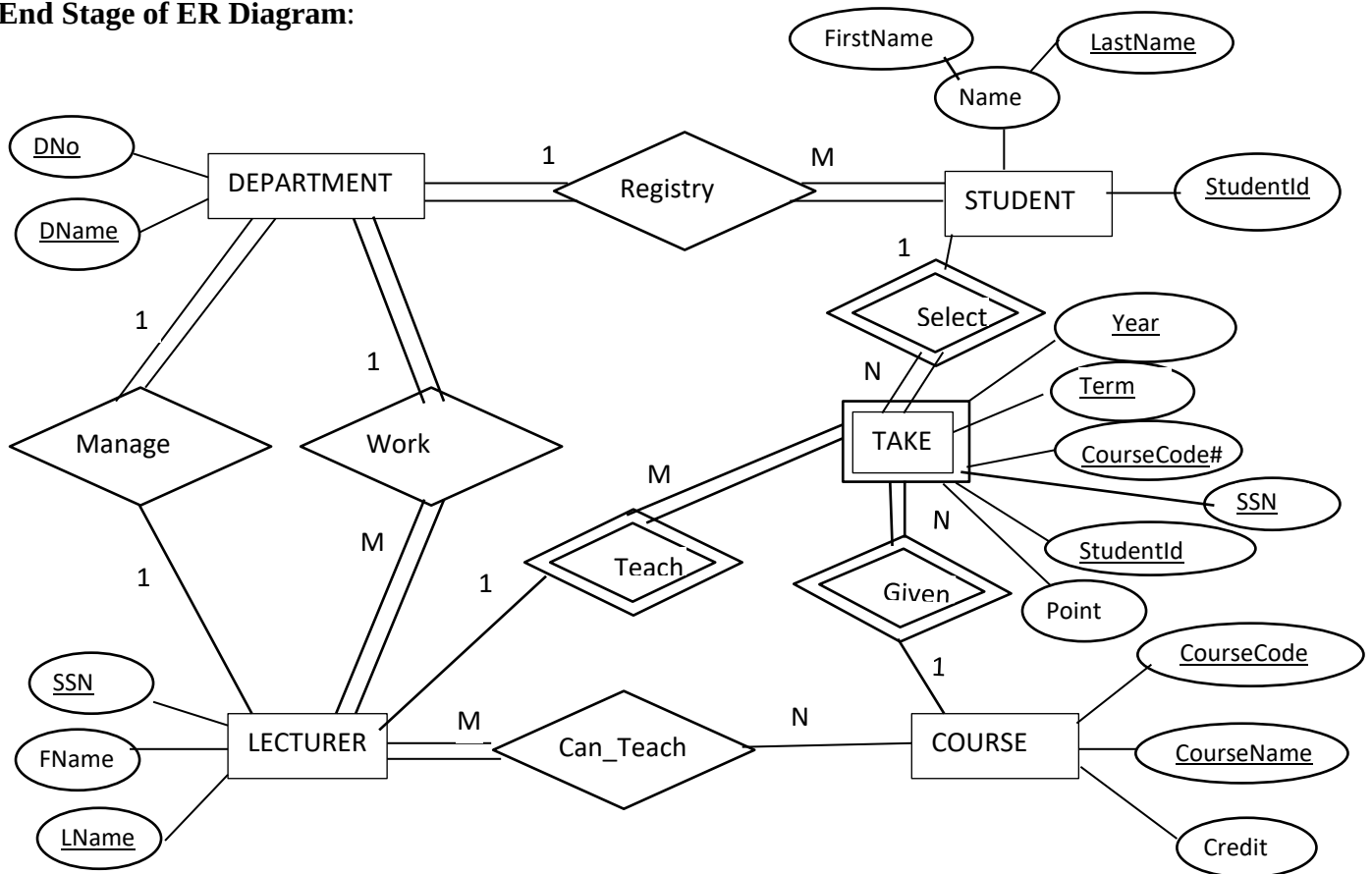
a) Start Stage of the ER Model



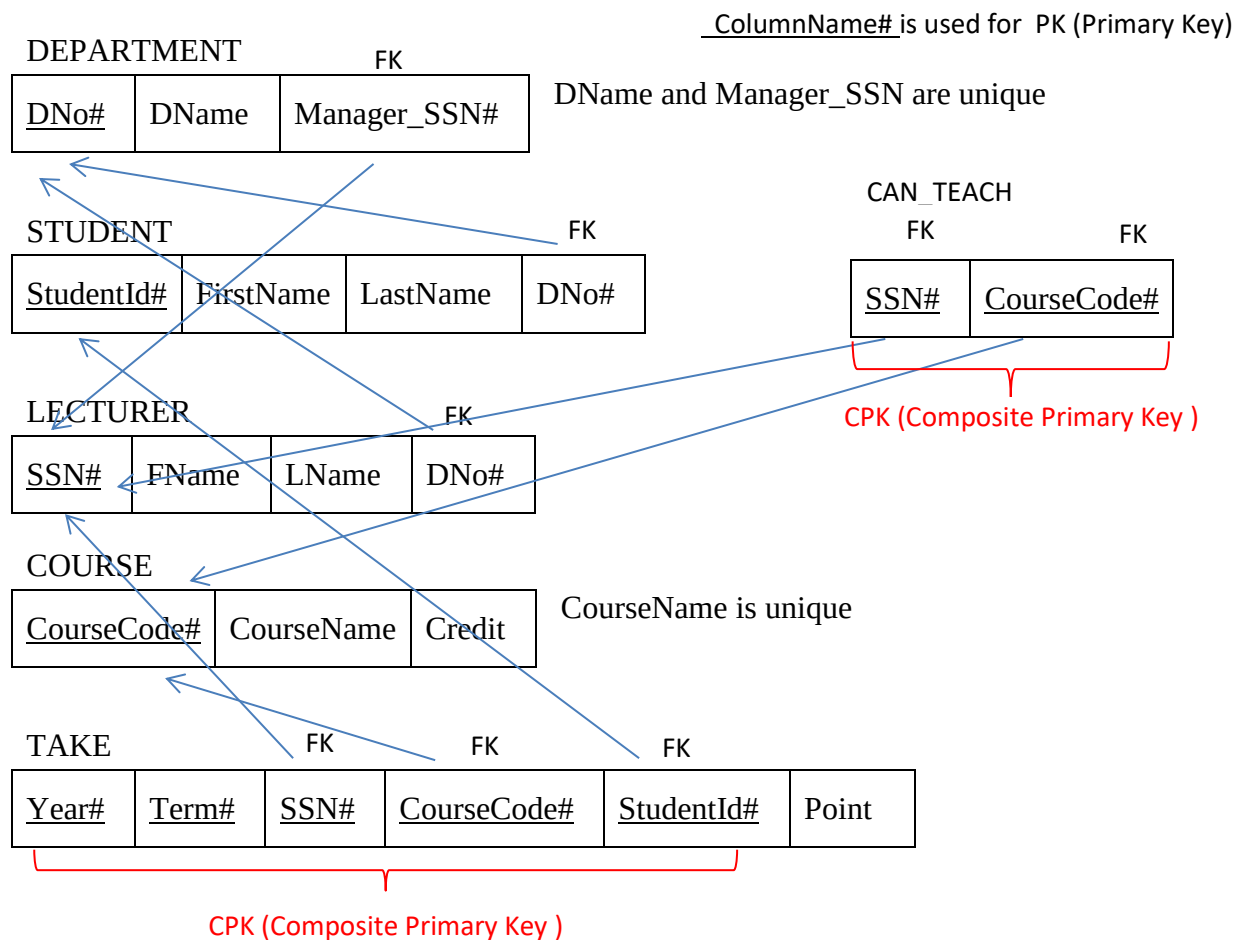
We can not decide about the relationship type of N-ary degree relationship such as Take, therefore we have to convert this relationship to weak entity having 3 binary relationships, which is shown in the following diagram which is the end stage of ER Model.

Exercise 1:

End Stage of ER Diagram:



b) Database Schema of Relational Model



Exercise 1:

c) Database schema's Diagram of the Relational Model

